

HOUSTON DUNN, TX, USA

WMO: 720594

Lat: 29.717N

Lon: 95.383W

Elev: 69

StdP: 100.50

Time Zone: -6.00 (NAC)

Period: 08-19

WBAN: 00188

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	0.0	2.4	-10.2	1.6	5.6	-7.9	1.9	8.4	11.0	10.6	10.1	12.0	4.7	350	0.421

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.2	37.7	25.0	37.0	24.7	35.1	24.6	26.4	32.4	26.2	31.9	25.8	31.1	2.1	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.2	20.6	27.7	25.1	20.4	27.7	24.9	20.1	27.7	82.6	31.7	81.2	32.1	80.0	30.8	27.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.6	8.5	7.7	-2.1	39.8	2.6	2.4	-4.0	41.5	-5.6	42.9	-7.1	44.3	-9.0	46.1		
			-4.5	27.2	2.4	0.5	-6.2	27.5	-7.6	27.9	-9.0	28.1	-10.7	28.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	22.3	12.9	15.3	18.4	21.8	25.5	29.1	30.1	30.8	28.0	23.6	17.8	14.2
	DBStd	7.36	5.37	5.68	4.61	3.29	2.73	1.93	1.68	1.76	2.40	4.04	5.02	5.35
	HDD10.0	85	34	20	4	0	0	0	0	0	0	6	21	
	HDD18.3	586	181	111	56	11	1	0	0	0	9	68	149	
	CDD10.0	4588	122	168	266	353	480	573	623	646	541	422	242	152
	CDD18.3	2046	11	26	59	114	222	323	365	388	291	173	53	22
	CDH23.3	20478	29	87	260	652	1858	3709	4446	4817	2934	1366	248	70
CDH26.7	9082	1	11	51	144	647	1766	2198	2509	1234	471	46	6	
Wind	WSAvg	3.4	3.8	4.2	4.2	4.3	3.7	3.1	2.5	2.4	2.7	3.1	3.7	3.8
Precipitation	PrecAvg	1085	82	66	63	74	113	128	92	102	121	85	81	75
	PrecMax	1580	170	146	156	254	280	381	210	270	378	283	182	180
	PrecMin	582	7	9	14	7	18	10	19	23	21	0	9	10
	PrecStd	253	46	35	36	63	64	109	43	65	74	72	52	45
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	26.0	27.9	30.0	32.0	34.1	37.9	37.9	41.0	37.8	33.9	30.1	27.5
		MCWB	18.5	19.8	20.5	19.4	22.4	24.6	25.0	25.4	24.4	23.9	22.8	20.5
	2%	DB	23.9	26.0	27.8	29.8	32.6	36.2	37.2	38.0	35.8	32.5	27.7	25.1
		MCWB	17.9	19.7	20.0	20.1	23.0	24.3	24.8	25.4	23.8	23.9	21.3	20.4
	5%	DB	22.4	23.9	26.2	28.0	31.3	34.9	35.9	37.2	33.9	31.1	26.1	23.7
		MCWB	17.6	19.0	19.2	20.1	22.9	24.3	24.6	25.2	23.9	23.3	20.5	20.3
	10%	DB	20.2	22.4	24.2	27.3	30.0	33.7	34.8	35.1	32.6	29.1	24.1	22.4
		MCWB	17.1	18.7	18.4	19.9	22.5	24.4	24.7	24.9	23.8	21.7	20.0	19.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.5	22.3	22.5	24.1	25.5	26.8	26.7	26.9	26.3	26.2	24.3	23.1
		MCDB	22.4	24.1	25.6	27.3	29.8	31.4	32.0	33.0	32.0	29.3	27.2	24.4
	2%	WB	20.3	21.4	21.9	23.4	25.0	26.2	26.2	26.6	25.8	25.6	23.3	22.3
		MCDB	22.0	23.5	25.2	26.2	29.1	31.1	31.7	33.1	30.8	29.0	25.7	23.9
	5%	WB	19.4	20.8	21.3	22.8	24.5	25.8	25.8	26.2	25.6	24.9	22.5	21.3
		MCDB	21.3	22.9	24.1	25.5	28.6	30.6	31.0	32.8	30.3	28.4	24.9	23.2
	10%	WB	17.9	20.0	20.7	22.2	24.0	25.5	25.7	25.9	25.2	24.2	21.3	20.1
		MCDB	19.3	21.8	23.1	25.0	27.9	30.2	30.7	31.8	29.5	27.7	23.9	21.7
Mean Daily Temperature Range	5% DB	MDBR	9.1	8.7	8.9	9.0	8.7	9.3	9.4	10.2	8.8	9.6	9.0	8.4
		MCDBR	10.3	9.3	9.9	9.9	9.8	10.6	10.6	12.1	10.6	9.8	9.1	9.3
		MCWBR	6.3	5.3	4.6	4.1	2.9	2.6	2.2	2.5	2.8	3.3	4.7	6.2
	5% WB	MCDBR	8.9	8.5	8.5	7.9	8.1	9.1	9.4	10.5	8.8	8.7	8.7	9.2
		MCWBR	7.4	6.4	5.1	4.5	2.8	2.5	2.3	2.5	2.6	4.4	5.3	7.5
Clear-Sky Solar Irradiance	taub	0.339	0.357	0.371	0.435	0.478	0.468	0.495	0.483	0.453	0.378	0.369	0.347	
	taud	2.505	2.464	2.416	2.237	2.153	2.234	2.179	2.214	2.263	2.464	2.438	2.489	
	Ebn at Noon	896	908	915	863	824	827	803	811	824	874	852	867	
	Edh at Noon	90	102	114	141	154	141	149	142	131	101	96	88	
All-Sky Solar Radiation	RadAvg	2.88	3.24	4.43	5.30	5.89	6.12	5.99	5.74	4.97	4.33	3.26	2.55	
	RadStd	0.28	0.56	0.38	0.38	0.43	0.50	0.56	0.47	0.52	0.52	0.33	0.31	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (35 neighbors)	+0.33	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+111	+97

Nomenclature: See separate page